

Tactile-Based Object Recognition Using a Grasp-Centric Exploration

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Abstract—As humans, our grasping and manipulation skills are highly dependent on our ability to perceive tactile properties. Conversely, most of today's robotic operations still rely predominantly on visual feedback for identifying the objects that need to be grasped and handled. In this work, we study the problem of recognizing everyday objects based solely on their tactile attributes. This has a significant practical value, as it could allow object identification even when visual sensing is impossible, or assist vision in difficult contexts. Our method consists of acquiring multi-modal tactile sensing data during a quick grasp-centric exploration phase, with minimal operational cost. Our algorithm was able to recognize objects from a considerably-large set of 50 general purpose items with an accuracy of 98.1%. Moreover, we show that it is possible to reliably identify a large proportion of these objects by only analyzing the deformation pattern that they undergo during compression. Further, we study our method's ability to learn relevant tactile properties to classify new objects. We also share our tactile sensing database that contains various sensor data acquired from more than 1600 experiments, which was used for this work. Finally, we discuss the relative performance and role of each tactile modality for differentiating objects.

I. INTRODUCTION

In the February 2007 edition of the Scientific American [1], Bill Gates anticipated that robotics would soon undergo a boom similar to the one the computer experienced in the 80s. The first materialization of this prediction has come in the form of a new, fast growing branch of robotics called collaborative robots. These robots, born out of a popular research trend that looks to harmoniously incorporate both robots and humans in the same workspace, have the advantage of being not only simpler to program, but also safer for humans. These particular technical characteristics have allowed collaborative robots, which appeared on the market in 2010, to quickly conquer a sector where automation was still not widespread, namely that of small and medium-sized enterprises (SMEs). This brilliant departure, as evidenced by the statistics from the International Federation of Robotics, represents just the tip of the iceberg as there still remains many opportunities for automation in these types of companies. Unfortunately, current collaborative robots do not yet have the capacity to perform the majority of tasks found in SMEs, where the production environment designed for humans is much less structured than that of the large production lines. One of the biggest technical bottlenecks that limits the potential massive integration of these robots

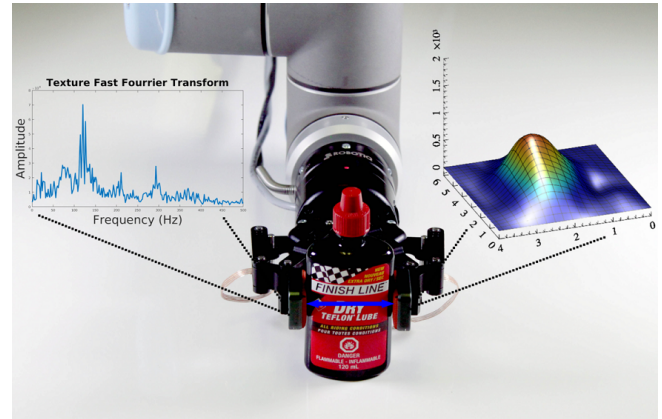


Fig. 1. A gripper using a set of tactile sensors to extract relevant tactile properties.

in this sector is their limited grasping capabilities, which are well below the level of a human. The commercial reality of SMEs, which often involves low-volume but high-diversity production, requires that any potential robots are able to reliably interact with a wide variety of parts and perform complex tasks such as assemblies.

In this new working paradigm, where robots have to grasp a multitude of different objects, being able to differentiate and recognize them becomes paramount. Several researchers have sought to solve this problem by using artificial vision, recent surveys on vision-based robotic grasping can be found in [2], [3]. An example of a vision-based robot is the well-known work presented in [4] that uses vision with deep reinforcement learning to perform grasping. However, even the current state of the art vision-based approaches suffer from technical limitations pertaining to cameras (e.g. lack of contrast, reflections, illumination, etc.), or task-related constraints (e.g. obstructed view by the gripper, translucent parts, cluttered environments, etc.) that introduce various uncertainties into vision-based operations. Moreover, while vision is good at mapping object geometry, it will generally fail to capture certain physical properties that could help identifying an object such as its stiffness, its surface texture, or its fragility. An example of this fact is shown in Fig. 2.

Rather than using artificial vision, the approach presented in this paper exploits tactile signals to recognize and differentiate objects. Recent works have demonstrated that the use of tactile signals allows the recognition of objects at a good success rate [5]–[7]. Being able to rapidly recognize objects using touch would be a valuable skill in contexts such as assembly lines where a finite set of parts are to be picked up and identified, for example, from a shelf or from a bin where

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they are mixed. This ability could potentially be used to assist or even replace vision in difficult contexts, for example when picking up parts in a cluttered and/or occluded environment, or where poor visibility prevails. In this vein, this work investigates the importance of different commonly-used tactile modalities for identifying typical everyday objects that were not designed or chosen specifically for the experiments. Particularly, the originality of this work lies in that it studies a variety of tactile feedbacks (proprioceptive, exteroceptive, static and dynamic) and discusses which one is better suited depending on the manipulation context. The contribution of each modality was evaluated on a large dataset encompassing over 50 objects. We deal with the advanced classification problem of automatically determining to which one of these 50 objects a specific, never-seen *tactile signature* belongs. Furthermore, while there are some examples of tactile-based object recognition in the literature, many of them rely on a more or less exhaustive tactile exploration phase. Thus, an additional contribution of this work is to propose a quick, grasp-centric exploration phase. Depending on the number and the nature of the objects to identify, we show that our approach only requires tactile sensing data to be acquired during a typical object grasp, hence generating little-to-no additional operational cost.

This paper adheres to the following structure. First, relevant literature on tactile sensing is introduced, with a particular focus on object recognition. Subsequently, the experimental approach implemented in this paper to study different modalities is outlined. Then, the experimental results are presented and analyzed. Finally, the significance of the results as well as future work they inspire is discussed.

II. RELATED WORK

A. Tactile Sensing in Robotics

Integrating tactile sensors to enhance robotic operations is far from being a novel concept [8]. Typically, the purpose of a tactile sensor is to provide content-rich and task-relevant information about tactile properties at the contact interfaces. Humans perform this feat using a dense array of biological mechanoreceptors, inherent to our sense of touch, to feel different tactile modalities. The most predominant modalities used during manipulation tasks are normal pressure, vibration, and shear sensing [9], [10]. The development of a wide variety of tactile sensors just over the last years demonstrates the general effort towards incorporating these types of feedback into robots. Using different tactile sensing technologies (which include optical, resistive, magnetic, barometric, and capacitive-based transducers), there have been various attempts to build a sensor capable of measuring one or many of these modalities. General reviews of tactile sensing technologies can be found in [11], [12]. In this work, we use the highly-sensitive multi-modal capacitive tactile sensor described in [13]. It is further discussed in the *experimental setup* subsection. This particular sensor allows us to study the performance of different modalities for the purpose of recognizing objects given only tactile inputs, which is in fact a basic human ability [14].



Fig. 2. Visually-similar examples of different Starbucks mugs that have different tactile properties. The left mug is made of ceramic, the middle one is made of cardboard and the right one is made of plastic. All mugs could be either full or empty, which would also change their tactile properties.

In recent years, tactile sensors have increased in popularity and are currently used in a wide variety of applications. These range from providing information about an object's pose [15], an object's shape [16] or the contact force distribution [17], to executing complex tasks such as automatic classification of dynamic events [18], texture recognition [19], or even the impedance-control of robots based on tactile feedback [17]. This work focuses on the task of recognizing objects by identifying their principal tactile characteristics.

B. Using Tactile Sensors for Object Identification

The concept of identifying objects using tactile sensing is a longstanding topic of research. For instance, in 1984, Eric et al. [8] were among the firsts to study object identification using data from a tactile sensor. Twenty years after, Heidemann et al. [20] took advantage of the progress made in tactile sensing technologies and used time series of pressure profiles, obtained from a more advanced tactile sensor, to discriminate household objects. The approach, tested on seven objects, involved detecting a first contact, then exploring the surface of the object while maintaining a steady contact pressure. While the average classification success rate was high ($\approx 81\%$), the method was tested on a small set of objects and involved a relatively complex exploration phase. Conversely, Spiers et al. [21] showed that it is possible to extract several object features during a single grasp. Using such features, the authors were able to identify a small set of everyday objects (11 items) at high accuracy ($\approx 94\%$). While both of these last two works used *static data* (i.e. a pressure map acquired at a relatively low frequency), Chen et al. [5] rather measured vibrations at a high frequency (i.e.: *dynamic data*) and was able to accurately identify an object inside a container when it was shaken. This shows that *dynamic data* also contains information that can be used for the purpose of identifying an object. However, having to shake an object placed in a container could be problematic as it could potentially cause damage to the said object (e.g.: fragile objects) and has a significant time cost. Pezzementi et al. [6] proposed to use a set of two differently handcrafted descriptors to identify objects, the first was directly built from

the tactile pressure map while the second was synthesized from the top-low frequencies in the Fast-Fourier Transform (FFT) of the same signal. While the two descriptors are different, they were both handcrafted from the same source of information and were tested on five plastic letters, which might not reflect the wide variety of textures, rigidity, shapes and sizes, among other properties, that is normally found in household or industrial objects. More recently, Lin et al. [7] studied how a machine learning agent could learn to associate a tactile pressure map snapshot of a grasped object with its corresponding 2D image captured by a camera for object recognition. While the number of considered objects was higher (98), only a small subset of candidate objects (either 5 or 10) was presented at the time of classification with an accuracy ($\approx 24\%$ to 40% for the first guess) that could make this approach impracticable in an industrial context. Corcodel et al. [22] used an unsupervised approach where they successfully were able to determine if a grasped object corresponded to a previously encountered object or not.

While all of the aforementioned examples show interesting results, the literature still lacks examples where multiple tactile modalities (different sources of tactile sensing elements) are simultaneously employed to efficiently identify objects without having to rely on vision. Furthermore, most of the object recognition scenarios still involve a relatively time consuming and/or complex tactile exploration phase, with the accuracy capping or rapidly decreasing when the number of objects to recognize increases. Consequently, few approaches can be found where it is shown that the generated features can be used to learn the properties of a considerable quantity of arbitrary and non-specifically designed objects. In contrast with the literature, we propose a multi-modal approach utilizing two tactile sensors and a fast, grasp-oriented tactile exploration step. We quantify and compare the performances of different modalities with respect to their capacity to correctly recognize objects. We constructed and

share a database¹ from 1600 experiments conducted on over 50 objects, which we show can be used to infer tactile properties to never-seen objects. The data was used to train different algorithms using either a single or a combination of modalities.

III. THE APPROACH

A. Experimental Setup

The data for this work was acquired using the tactile sensor described in [13]. It measures the normal pressures at 60 Hz using 28 capacitive sensors organized as a 4×7 tixel pressure image. A standard integrated inertial measurement unit (a MPU-9250 IMU, i.e.: three accelerometers, three gyroscopes, and a compass) provides proprioceptive feedback at 1kHz (which is particularly important for an under-actuated gripper) and a transimpedance amplifier measures the instantaneous capacitance variation at 1kHz. We refer to the 4×7 tixel matrix as the *static data* and to both the instantaneous variation of capacitance and the accelerometer data as the *dynamic data*. A two-finger Robotiq© gripper was outfitted with two of these sensors, as shown in Fig. 1. The gripper was mounted on a UR10 robot from Universal Robots©. All the hardware was controlled and synchronized using ROS *Melodic* on Ubuntu 18.04.

Using this setup, we performed a total of 1641 tactile exploration experiments on 50 objects (half of them are the *Amazon Picking Challenge* objects [23] to which we added other random household objects: some are shown in Fig. 3), for an average of 33 experiments per object. 70% of this dataset was used for training (with 1/5 used for validation) and testing was done on the remaining 30%, with equal object representation in all sets. We also conducted additional experiments (30 per object) on five never-seen objects and two previously-seen objects whose surface texture was altered on purpose to study which learned features would be selected in these cases.

B. The Tactile Exploration Phase

Acquiring information using tactile sensors during a first-touch sequence should be simple, fast, and non-invasive while still providing enough information to infer rich features for the task at hand. Having these constraints in mind, we first began by investigating a three-step sequence (depicted in Fig. 1 and demonstrated in the accompanying video) to acquire data for each modality. As this work does not study grasp planning, the pose of the gripper relative to the object was assumed known and was arbitrarily chosen at different locations on the objects prior to the robot sequence. More precisely, the objects were first positioned on a table at a predetermined location with different orientations such that the gripper would evenly touch all their sides, excluding the side(s) larger than the gripper's stroke or those impossible to rub across one inch. Then, we apply the following procedure:

1. We move to the object's position (about one inch over the object's centroid, as shown in the accompanying



Fig. 3. Some of the 50 objects used during the experiments.

¹This 2.5Gb database is shared as part of this submission here: www.dropbox.com/s/wumz72yetmp9zg6/Experimental%20Data.zip?dl=0

video) and incrementally close the gripper until a first contact is detected. This is signified by the sum of the static taxels surpassing a given threshold value. We record the *opening of the gripper* right after the first contact to measure one of the object's dimensions.

2. Once a slight contact has been established between the object and the tactile sensors, we initiate a "rubbing phase", by moving the robot's end effector downward by approximately one inch. During this phase, we continuously record the tactile *dynamic signals* consisting of the sensor's accelerometer feedback, measured from the axis normal to the sensor's surface, and the instantaneous variation of the capacitance signal. The rationale behind this phase is to extract information that is a function of the object's surface properties, such as roughness and texture. If the object has a variable size over the distance to be traveled (e.g.: an irregular shape, a spherical shape, etc.), we still travel one inch downward and record the data, hypothesizing that the absence of vibrations at certain locations will nevertheless reveal information about the nature of the object.
3. Finally, the gripper is commanded to close at half of its default speed and force to explore the object's stiffness. During learning, the object is squeezed five times during this phase for data collection purposes (however it is only squeezed once during testing / actual object identification). For each squeeze we keep four static data snapshots as shown in Fig. 5(a) and 5(b). The idea is to capture the *deformation perceived at the fingertips*, which is essentially the evolution of the contact area and normal pressures during the compression.

C. The Machine Learning Agents

To extract relevant high-level features from the tactile signals, an appropriate algorithm and/or machine learning technique must be selected. One of the primary challenges of any pattern recognition task is dealing with the nature of the data and their variability. This is particularly true in the case of tactile sensors, as Qian et al. [24] demonstrated, since they often have a high variability in their feedback. In their work this resulted in unreliability when predicting grasp stability. Given these concerns, this section outlines our design of a robust agent capable of extracting pertinent information from the input signals.

1) *The Gripper Opening Position*: As described in the *tactile exploration phase* section, as soon as the initial contact is detected between the object and the tactile sensors, the gripper opening position is recorded. This value is an integer that can vary between 0 (completely open) and 255 (completely closed) and is derived from the encoder attached to the motor that drives the gripper. Given a single object that is grasped multiple times at the same location, the gripper opening value (G) will slightly fluctuate between experiments due to different phenomena (e.g.: the gripper's repeatability, the tactile sensor's repeatability, permanent

deformation in the object, etc.). Hence, here we assume G to be a random variable following an object-specific Gaussian distribution:

$$G_{i,k} \sim N_{i,k}(\mu_{i,k}, \sigma_{i,k}), \quad (1)$$

where $N_{i,k}$ is the assumed Gaussian distribution of object i touched on side k , where $k \in \{1, 2, 3\}$ is an index identifying the side where the object was touched. For example, if it was only possible to touch/rub the object on one of its sides, then $k := \{1\}$, while $k := \{1, 2, 3\}$ for an object touched on three of its sides. $\mu_{i,k}$ and $\sigma_{i,k}$ are respectively the average and the standard deviation of the gripper opening at the time of initial contact with object i on side k , where $i = \{1, 2, \dots, 50\}$. During training, it is thus possible to characterize $\mu_{i,k}$ and $\sigma_{i,k}$ for each object/side. After the training phase, we can then use these distributions to classify a gripper opening G by determining which class C is the most likely to correspond to the input G , yielding the highest probability for G :

$$C = \arg \max_i \left[\frac{1}{\sqrt{2\pi\sigma_{i,k}^2}} e^{-\frac{(G-\mu_{i,k})^2}{2\sigma_{i,k}^2}} \right]. \quad (2)$$

2) *The Dynamic Data*: The underlying hypothesis for this section is that the frequency component of a tactile sensor's dynamic signal will vary according to the object's surface properties. Hence, the first step for using this signal for classification was to convert it to the frequency domain using a regular Fast Fourier Transform (FFT).

In a previous work [18], performed with another tactile sensor, sparse coding was put into contribution to build a dictionary of 128 high-level features that was employed to successfully classify dynamic events at a high accuracy. The concept was to build a dictionary based on the most important features (from a signal reconstruction point of view) of a dynamic signal's FFT. However, the question of whether this dictionary could generalize to other sensors and/or tasks remained. Therefore, in this work we compared the classification success rate obtained by using a standard logistic regression on the FFT, to the success rate obtained using this previous dictionary of features. For the latter case, the dynamic signal's FFT was sparsely encoded using the previously-learned dictionary and the resulting sparse vectors were fed to a simple fully-connected neuron layer, as shown in Fig. 4.

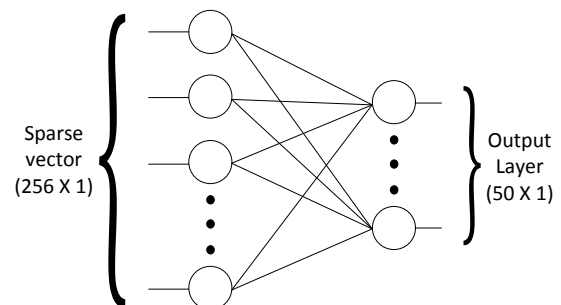
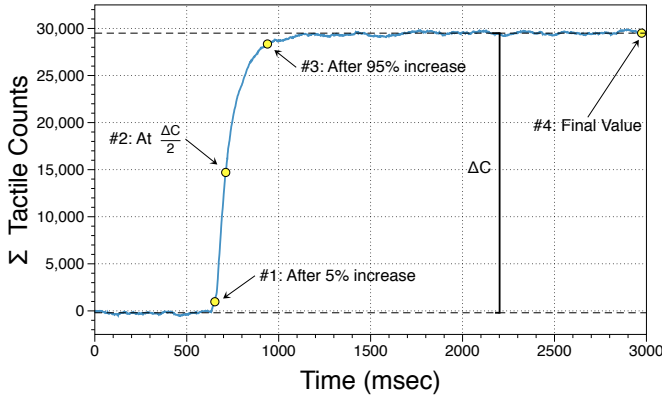
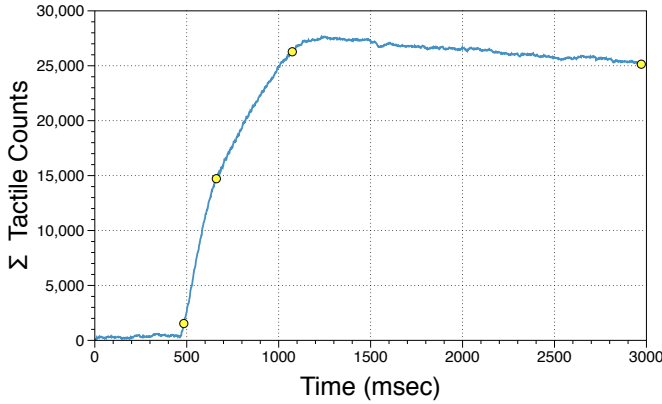


Fig. 4. Two sparse vectors (128 elements each) are concatenated and linked to a simple fully-connected layer with 50 output nodes, where each node correspond to a specific object. Here, the 128 elements are due to the 128 features contained in the dictionary (see [18]).



(a) Four different times (yellow dots) are considered for static data selection.



(b) An example where an object (here, a shampoo bottle), has continued to deform even after the squeeze, decreasing the tactile count sum.

Fig. 5. Times at which tactile data was acquired

3) The Perceived Tactile Deformation at the Fingertips:

The way an object deforms under compression is a tactile property influenced by its geometry and stiffness, among others. The tactile data generated during this compression phase obviously contains a lot of meaningful information, since both the object's footprint on the fingertips and the normal pressure are evaluated at different moments and might be closely-related to each of the different considered objects. Also, it is worth noting that our tactile sensors internally use a smooth dielectric providing it some degree of compliance too. Thus, even in the case of very rigid objects (e.g., rocks), the sensors will slightly adapt to the rigid object's local shape as the grasping force increases, again revealing some object-specific features. During each squeeze, four tactile pressure maps (i.e. the concatenated *static* data of both tactile sensors) are taken at different times corresponding to: 1) when the sum of all the taxels reach 5% of their maximum value; 2) when the sum has reached half of the maximum value; 3) when the sum has reached 95% of the maximum value and 4) 3 seconds after the beginning of the squeezing process (see Fig. 5). The first three snapshots encapsulate the evolution of the object's deformation (i.e. how the contact areas and the magnitudes of the normal pressures change) during the compression. Afterward, the last snapshot provides information about whether the internal forces within the grasp continue to change even after the gripper stopped moving. Fig. 5(a) is an example of a rigid

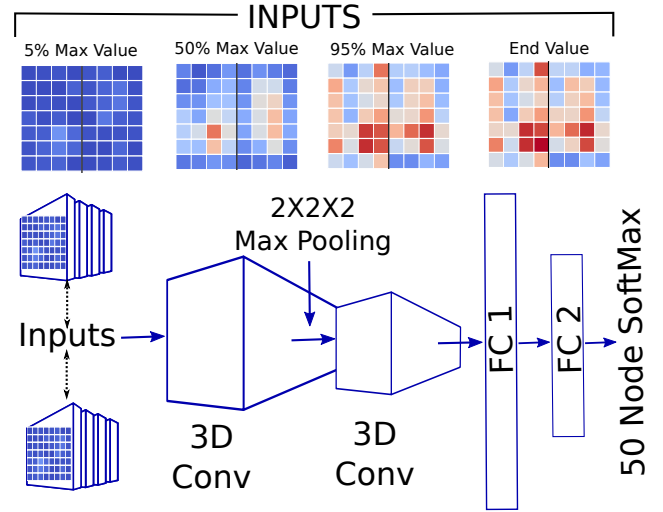


Fig. 6. The 3D convolutional neural network used to process the perceived deformation at the fingertips

object preserving its shape after the gripper has stopped its motion while Fig. 5(b) shows a softer object (a shampoo bottle containing a viscous fluid) that continues to change after the active compression is stopped.

To benefit from both the object's footprint shape and the pressure maps information, the static data from each sensor were combined to create a side-by-side 7x8 taxel image at each of the four aforementioned instances. These pressure maps were then stacked to create a three-dimensional spatiotemporal input that was fed into the three-dimensional convolutional neural network shown in Fig. 6. The network extracts features that maintain their spatial and temporal order which helps to classify objects based on their reaction to compression.

IV. EXPERIMENTAL RESULTS AND ANALYZES

A. The Contribution of Each Modality

The tactile modalities described in the approach section were tested both independently and in conjunction with each other to evaluate their respective contributions to touch-based object recognition. To fuse different modalities together, we simply concatenated their respective outputs and connected that to two fully connected layers that were trained separately. The final results are presented in Table I.

The top section of Table I shows the classification rate when only one modality is involved in the process. By only squeezing the objects, we were able to achieve a classification rate of 97.7% on 50 objects. This suggests that the information embedded in the progression of an object deformation contains high level and unique features that can accurately characterize an item from the set. Thus, it appears that an object's deformation-related properties are more pertinent to object recognition than the other modalities.

To further analyze this result, we used t-Distributed Stochastic Neighbor Embedding (t-SNE) [25] to visualize the compression data in a 2-D space. Fig. 7 shows the results obtained by applying this technique. One can clearly observe that the deformation data inherent to each object seems to gather in easily-distinguishable groups, which seems to

TABLE I
CLASSIFICATION RATE OF EACH MODALITY COMBINATION

Modalities	Classification Success Rate
Gripper Opening	67.1%
Dynamic Data ²	38.1%
Dynamic Data ³	48.6%
Perceived Object Deformation	97.7%
Opening + Dynamic ³	86.6%
Opening + Deformation	98.1%
Dynamic ³ + Deformation	97.8%
All together	98.1%

further confirm that each object from our dataset produced an almost-unique deformation pattern. This might be partly attributable to the complex but specific interactions between each single grasped object and our mildly-soft tactile sensor. Also, it might be relevant to note that there was a particularly large number of data acquired for this modality alone. Indeed, during the data collection process, we performed 5 squeezes per experiment (see the shared database), resulting in more than 6000 deformation patterns for the 50 objects.

Using only the gripper opening position achieved a 67.1% classification accuracy. This is likely the result of several factors such as groups of objects with similar dimensions, the variance in the tactile and gripper signals, and finally the discrete nature of the gripper command introducing gaps between the possible gripper positions.

The lowest classification rates were obtained using the dynamic modality, which could be the symptom of several factors. For example, most of the objects either had a smooth surface or a similar surface roughness. This, coupled with the fact that our tactile sensors might not have a sensitive-enough dynamic modality could explain this lower classification success rate. On the other hand, keeping in mind that randomly classifying objects would lead to an expected success rate of 2% ($=1/50 \times 100\%$), it is evident that dynamic data still contain a significant level of relevant information.

An interesting result to note is the important improvement in the classification rate when fusing the dynamic data with

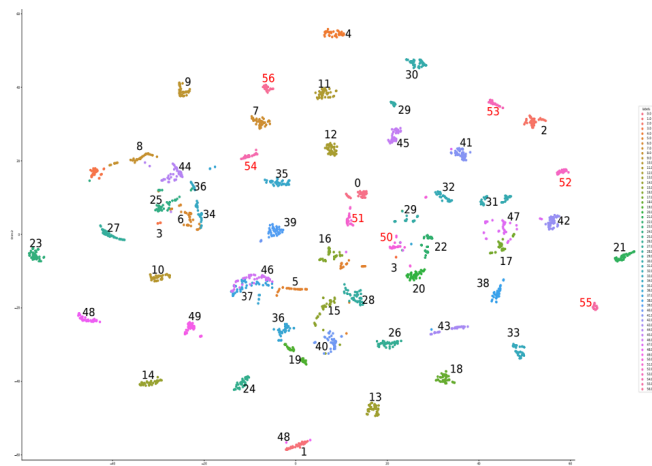


Fig. 7. t-Distributed Stochastic Neighbor Embedding (t-SNE)

²Using logistic regression

³Using sparse codes with a fully-connected layer

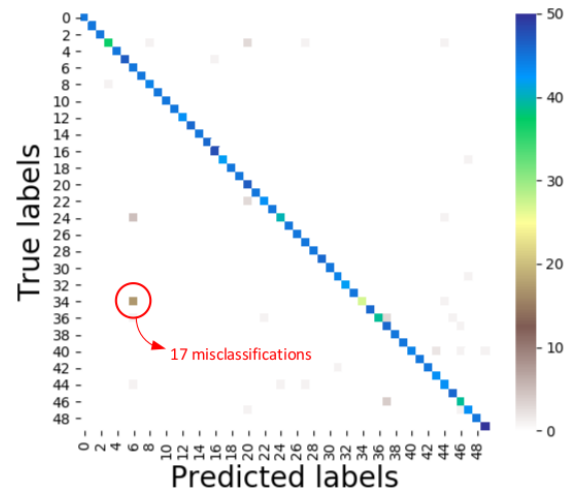


Fig. 8. Confusion matrix when all modalities are used in synergy

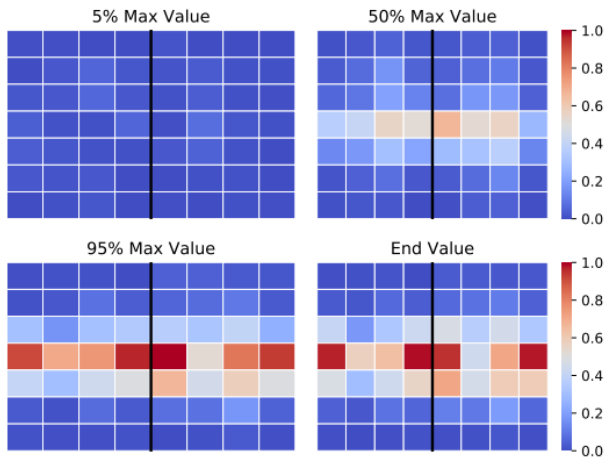
the gripper opening position as compared to using each of them independently. This would be useful in a case such as manipulating a fragile object, where the invasive squeezing phase is more likely to damage the objects. In this scenario, it would still be possible to achieve a respectable classification rate of 86.6%. The other combinations of modalities only slightly improve upon the results from the perceived deformation alone.

B. Sources of Confusion

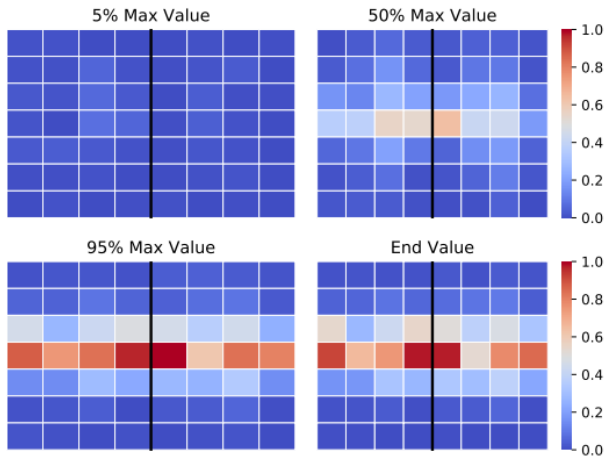
The classification results show that our machine learning agents can classify most of the objects at high accuracy, however there are some rare exceptions. Fig. 8 depicts the sources of confusion in each class. As expected, most of the tested objects are accurately classified, except in one case where there were 17 misclassifications which is significantly higher than the average of the other classes. This confusion happens between the can of peas and the coconut water bottle, as shown in Fig. 9. While this high number of misclassifications seemed surprising at first, touching the objects ourselves and



Fig. 9. Two objects that were confused: a can of peas (shown on the left, its label was removed to better show the surface properties) and a coconut water bottle. Conversely to Fig. 2, these visually-different objects are similar from a tactile sensing perspective: they are both very rigid and both have a smooth but corrugated surface. The perceived deformation patterns look alike and so does the surface properties (rubbing phase - dynamic data).



(a) Tactile map evolution while squeezing the can of peas.



(b) Tactile map while squeezing the coconut water bottle.

Fig. 10. Tactile maps during the squeezing phase of two confused objects.

removing the thin paper label from the can of peas gave a strong probable cause for this higher undesirable rate. Indeed, both objects are very rigid and do not deform much under pressure, but most of all: removing the label revealed they also seem to have an almost-identical corrugated surface. This corrugated surface is likely to create a similar dynamic signal when rubbed, since the grooves alternate in the same way. Most importantly, depending on where they are touched, the two objects are very likely to produce a similar footprint and deformation pattern at the fingertips. While such misclassifications are undesirable, these two objects are very similar from a tactile sensing perspective. Fig. 10 shows their similar deformation behaviors. While this makes them difficult to differentiate, one could argue that, given their similar tactile signatures, both objects could be grasped using a similar strategy. There are analogous problematic cases in vision-based object recognition where two objects are visually very similar but have vastly different tactile properties, for example the three coffee cups in Fig. 2. However, in these latter cases, the objects would often need to be grasped using distinct strategies: for instance, a ceramic cup is very rigid but slippery, while a cardboard cup has a higher friction coefficient but is very deformable and cannot sustain the same level of normal grasping pressure.

C. Never-Seen Objects and Property Inference

To investigate on how well the previously-described machine learning agents would individually deal with novel objects, we experimented on both never-seen objects and modified already-seen objects. Table II shows how these objects are classified using a single modality at a time, with the two highest scoring classes being reported in the column on the right.

The first object that was presented was a new kind of wood block (Maple) of smaller dimensions than the one used in our previous experiments. Also, Maple is a hardwood while the previous wood block that was used was made of white pine, a softwood. Interestingly enough, that didn't stop the perceived deformation network from properly classifying this object most of the time (75.2%). Unsurprisingly, it was also properly classified by the dynamic data classifier, since the surface finishes were similar.

The "Wood + Net" object is simply the original wood block, but this time covered with a thin layer of a net (from a previously seen object: see the *orange net* from our shared database) so that only its texture would change. As expected, the squeezing process still allowed a good classification rate for this scenario (69.3%), and the dynamic classifier was able to detect the texture at a high success rate. In fact, the netting's texture is so particular that it produced a classification rate of 80.8%. Similarly, we also covered the aluminum part (previously seen) with sandpaper (also a previously seen object: see the *sandpaper medium 100* object). The dynamic data classifier made good predictions (it was right 89.6% of the time). On the other hand, this time the static data classifier has assigned a wrong object 90% of the time. This misclassification might indicate that covering the aluminum part with sandpaper slightly changed its deformation pattern. Indeed, the "board" is a flat cutting board that has a texture similar to the sandpaper and which is very rigid. Despite this wrong assignation, an interesting fact is that we can still infer that the squeezed object was flat and rigid. In fact, for the majority of the misclassified static cases, the object was still mapped to one that had similar tactile properties, implying that the result could be useful for future work, for example, on grasp planning and manipulation.

Finally, two new and never-seen objects (a perfume and a cough syrup bottle) were introduced to our classifiers. The

TABLE II
CLASSIFICATION OF NEVER SEEN OBJECTS

Unseen Objects	Modality	Classification
New Wood	Static	75.2% (Wood), 20.0% (Stone)
New Wood	Dynamic	70.4% (Wood), 18.9% (Spoon)
New Wood + Net	Static	69.3% (Wood), 13.3% (Fabric)
New Wood + Net	Dynamic	80.8% (Net), 6.7% (Fabric),
Alu + S. P.	Static	90.0% (Board), 6.7% (Alu)
Alu + S. P.	Dynamic	89.6% (S.P.) 5.6%(Case)
Perfume	Static	71.0% (Wood log), 25.8% (DVD)
Perfume	Dynamic	59.2% (DVD), 20.0% (Wood)
Cough Syrup	Static	100.0% (Salt)
Cough Syrup	Dynamic	47.2% (Container), 33.8% (Salt)

first is a rigid and flat bottle made of glass with a wood frame. The second is a plastic bottle with medium rigidity. Although these objects were never memorized by our machine learning agents, the way they are classified reveals some of their tactile properties. For example, using only the classification results to infer tactile properties, one could have guessed that the perfume is likely to be a flat and rigid object, since both the wood log (which is basically another flattened piece of wood with a different texture) and the aluminum were both flat and rigid objects that didn't change their shape much during compression. It is also expected to have a smooth texture like that of the DVD case (whose texture was similar to glass when felt by a human). Similarly, the cough syrup bottle was mapped to the only other plastic bottles (i.e.: the plastic container and the pink salt) from our dataset.

V. CONCLUSION

As originally hypothesized, tactile sensors can be used to measure important tactile modalities such as perceived deformation, texture, and geometry which in turn can be used to recognize a wide variety of objects. The best result of this work was being able to identify objects with an accuracy of 98.1% using a combination of all the measured tactile signals. By far, the most useful information to properly recognize the considered objects was the perceived deformation at the fingertips during compression. Using only this information, it was possible to correctly classify the objects 97.7% of the time. Given this very high level of reliability, we propose to potentially only rely on this information to identify an object by recording the tactile pressure maps during grasping (see the accompanying video) at no additional production cost. Grasping an object using a significant level of force (half of the gripper's max force) seem to exert enough compression to capture important object-related properties. The fact that the capacitive tactile sensor we used had a slightly compliant dielectric might also have helped in some cases, since it could have been able to locally comply with the shape of rigid objects.

We also demonstrated that the proposed algorithms could be used to infer probable tactile properties of novel objects by matching them to similar objects that have been previously seen. In future works, we will explore how the extracted features could be automatically gathered in groups of relevant characteristics for grasp planning and object manipulation. For example, depending on its weight, a deformable object with a very smooth texture will not require the same grasping strategy as a rigid one with a coarse texture. A robot should be able to adapt to these differences.

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